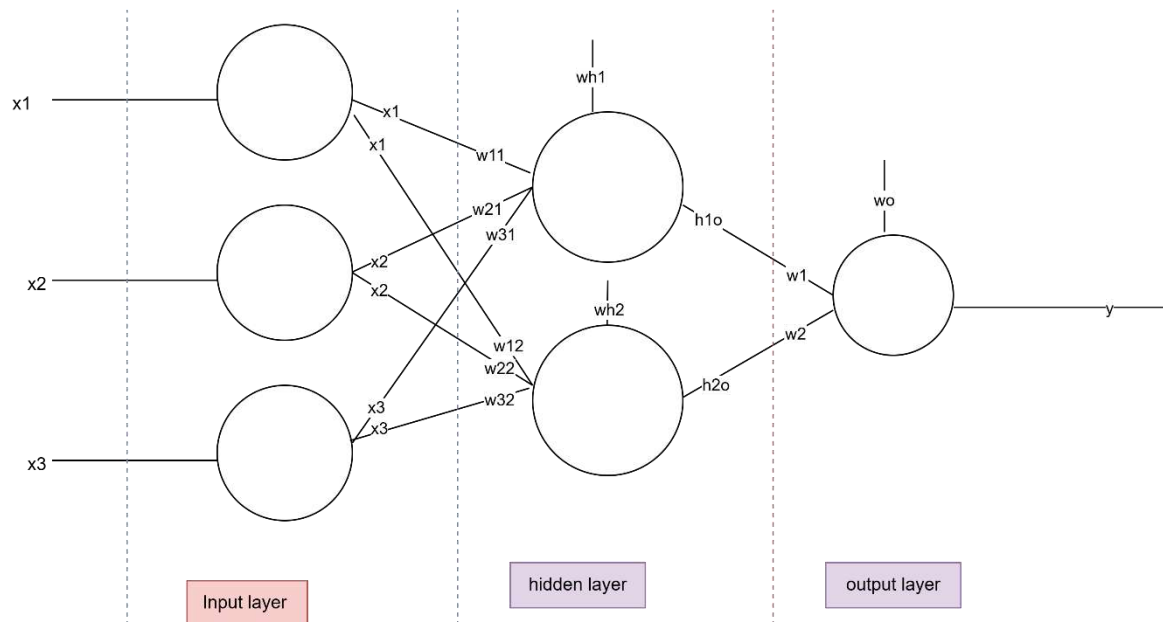


Deep learning

1. We are able to draw the NN
2. We are able to write math equations
3. we are able to write math equations in matrix format
4. we are able to calculate total parameters of any NN
5. we are able to calculate the parameters
 - Forward propagation
 - Backward propagation
 - Epochs
 - Gradient descent algorithm



Types of Gradient descent

1. Batch Gradient descent
2. Mini Batch Gradient descent
3. Stochastic Gradient descent

1. Batch Gradient descent

- It involves all the training datasets to compute the Gradients to update the model parameters
- For example assume that a training dataset has 1000 observations are there
- all 1000 observations are used to update the model parameters
- this one time updation means one epoch completed
- For next epoch these training observations calculate the error or gradients by using of latest parameters which is updated in previous epoch

always remember one iteration or one epoch means all the observations need to

travel along the NN, while traveling back to front (Back propagation) they will update the weights

but here the final updated weights can be used after all the observations iteration completed

	$C1$	$C2$	$C3$	$C4$	$C5$	y_{actual}	error
1	5	7					
2	10	72					
3	20	89					
1000	200	500					

How many input neurons required = 5 Neurons

every time one observation : $[c1, c2, c3, c4, c5]$ ===== y_{pred} vs y_{actual} ==
= error (Gradient) ===== Forward propagation

BP: Gradient Descent ===== update the weights

the above process will apply for all 1000 observations

then the final parameters available for usage

2. mini Batch Gradient descent

- here the observations will divide into Batches, the model parameters update by
- batch wise, which means the next batch can use the updated parameters
- but one iteration or one epoch wil complete after all the batch process completed
- here for every batch we will get parameter updation
- generally batch size 32,64,128 etc
- for example 1000 observations
- if we choose 64 batch size
- the number of batches $\frac{1000}{64} = 15$ (aprox)
- 15 times weights will update for every epoch
- if dont choose any size by default python give : 32

3. Stochastic Batch Gradient descent

- the model parameters update by each observation
- the next sample can use the latest paramters updated by previous sample
- but one iteration or one epoch wil complete after all the sample process completed
- here for every observation, we will get parameter updation
- for example 1000 observations
- 1000 times weights will update for every epoch

	<i>Batch</i>	<i>Min batch(10Bs</i>	<i>Stochastic</i>
<i>Data usage</i>	<i>All the data</i>	<i>sample or batch</i>	<i>single</i>
<i>Speed</i>	<i>low</i>	<i>Moderate</i>	<i>Fast</i>
<i>Memory</i>	<i>high</i>	<i>Moderate</i>	<i>Low</i>
<i>Parameter upation(1000 obseravtion)</i>	<i>one time</i>	<i>100 times</i>	<i>1000times</i>
<i>Time calculation per epoch(each observation 1s</i>	<i>1000sec</i>	<i>1000sec</i>	<i>1000sec</i>
	<i>Accuracy good but speed and memory draw back</i>	<i>Accuray and Speed both compromise</i>	<i>Speed and memory good but convergen ce noisy</i>
<i>Useful for</i>	<i>Less data</i>	<i>General use</i>	<i>Real time applicatio ns (end point prediction s)</i>

